INVEST WITH

WIND TURBINES

1 Jan - 1 Dec 2024

Maryam Ahmed
Mithaq Alhayki
Abdullah Zayed

DAB 24

INTRODUCTION

Prior to the expansion of wind energy, the United States **relied primarily on oil and natural gas for its power generation.** Although early turbine development dates back to the **1880s**, commercial utility-scale wind power truly accelerated in the **1980s**.

Today, **wind infrastructure** spans **88% of U.S**. **states (44 out of 50 states).** The technology operates as wind turns **rotor blades** around a **hub**, spinning **an internal generator** that delivers **electricity** directly to the **power grid.** Evaluated by **nameplate capacity (MW),** these **assets** provide a **clean**, **reliable**, **and renewable energy source** for local **communities** across the nation.

OBJECTIVES

1. Market Overview
Objective: Deliver a clear picture of electricity generation by measuring total output and fuel contributions, tracking performance trends over time, and assessing the renewable versus fossil fuel share in the overall mix.

2. Wind Energy Contribution
Objective: Isolate wind energy's specific generation share from total generation

3. Turbine Deployment Overview
Objective: Map the physical distribution of wind turbines across states and regions to identify key wind corridors.

4. Turbine Quantity vs. Net Generation
Objective: Evaluate whether turbine count per plant is the primary determinant of total net power output.

OBJECTIVES

5. OEM Manufacturer & Model Outcomes
Objective: Benchmark operational outcomes, reliability, and technical efficiency across turbine manufacturers and models.

6. Observed Turbine Capacity Performance
Objective: Assess actual turbine performance by calculating observed capacity factors against turbine performance.

7. Total Financial Sales & Revenue
Objective: Measure total electricity sales and revenue generated for power plants.

SCENARIO

Wow Company is exploring investment in a **U.S. industrial electricity generation project**, with a particular focus **on wind power** targeting **250 MW/month** to achieve **$20M–$25M** in monthly sales. To guide their decision, we will analyse the **2024 performance data of power plants across different locations**, comparing variations in **wind turbines, highlighting the leading turbine manufacturers, models, technical specifications** and **operational outcomes**. .

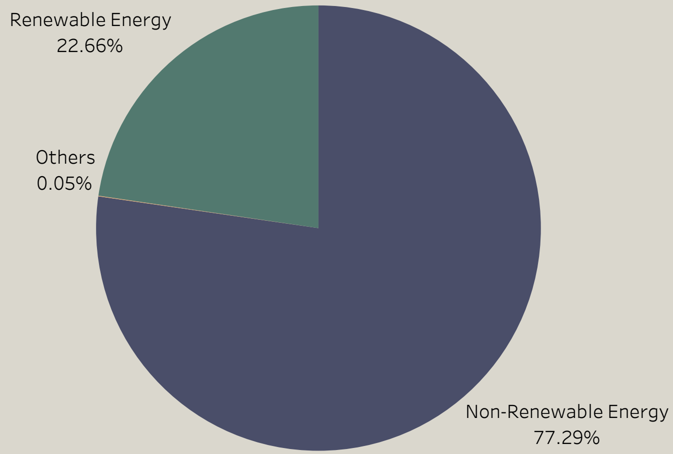
Total Net Generation

4.31B

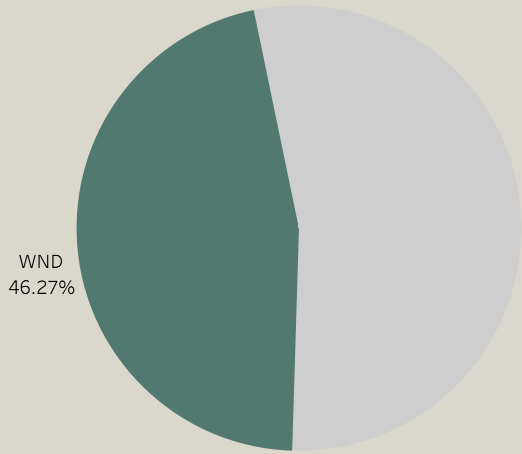
Number of Fuel Types

42

Objectives - A | Objectives - B | Scenario | Overview Summary | Fuel Types Contribution to Net Generation | Wind Contribution to Renewable Net Genera.. | Over time Net Generation per Fuel Type | Wind Turbines Distribution - Top 5 Sta.. | Number of Turbines & Net ..

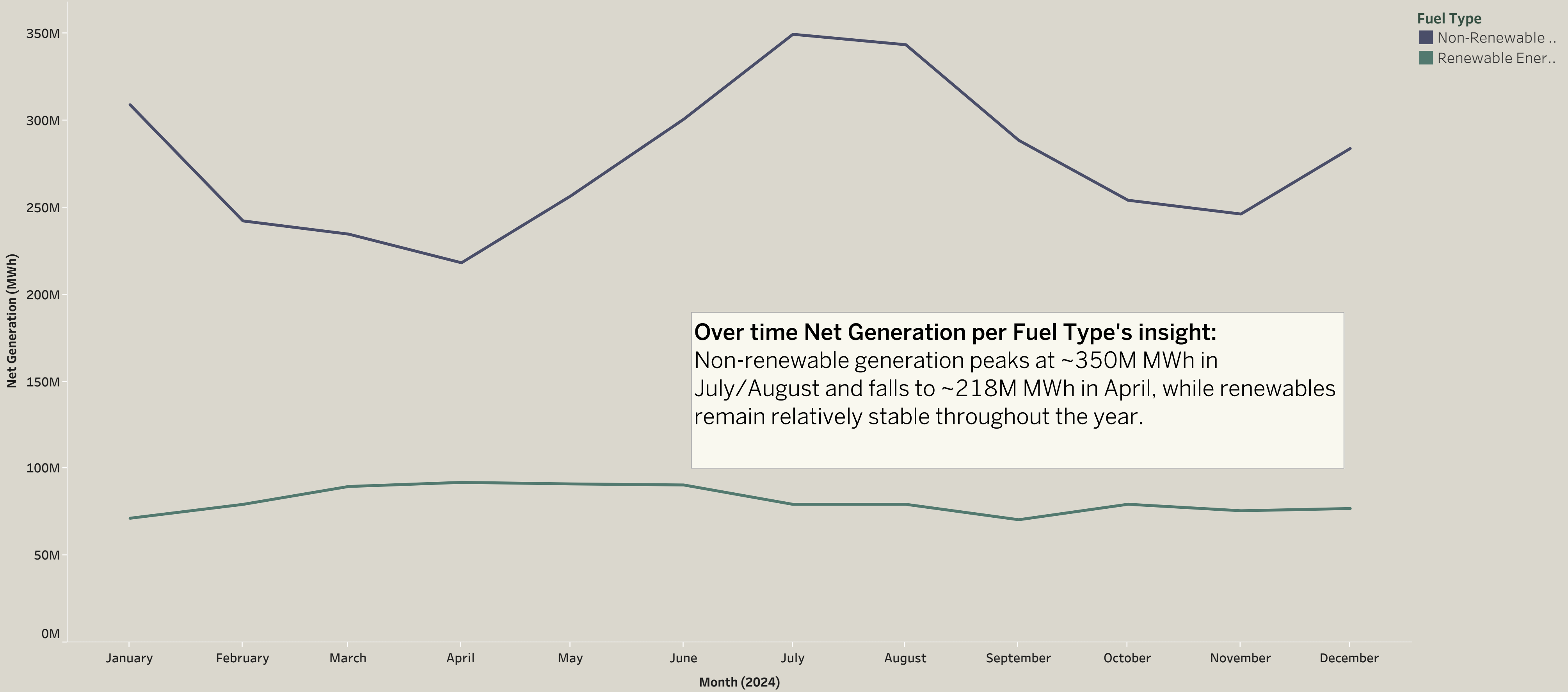


Fossil fuels remain a major source of electricity in the US because they can **generate power on demand** and provide a **reliable supply when electricity demand changes**. The country's **existing fossil-fuel infrastructure** also makes it **easier to maintain** their important role in the power grid.



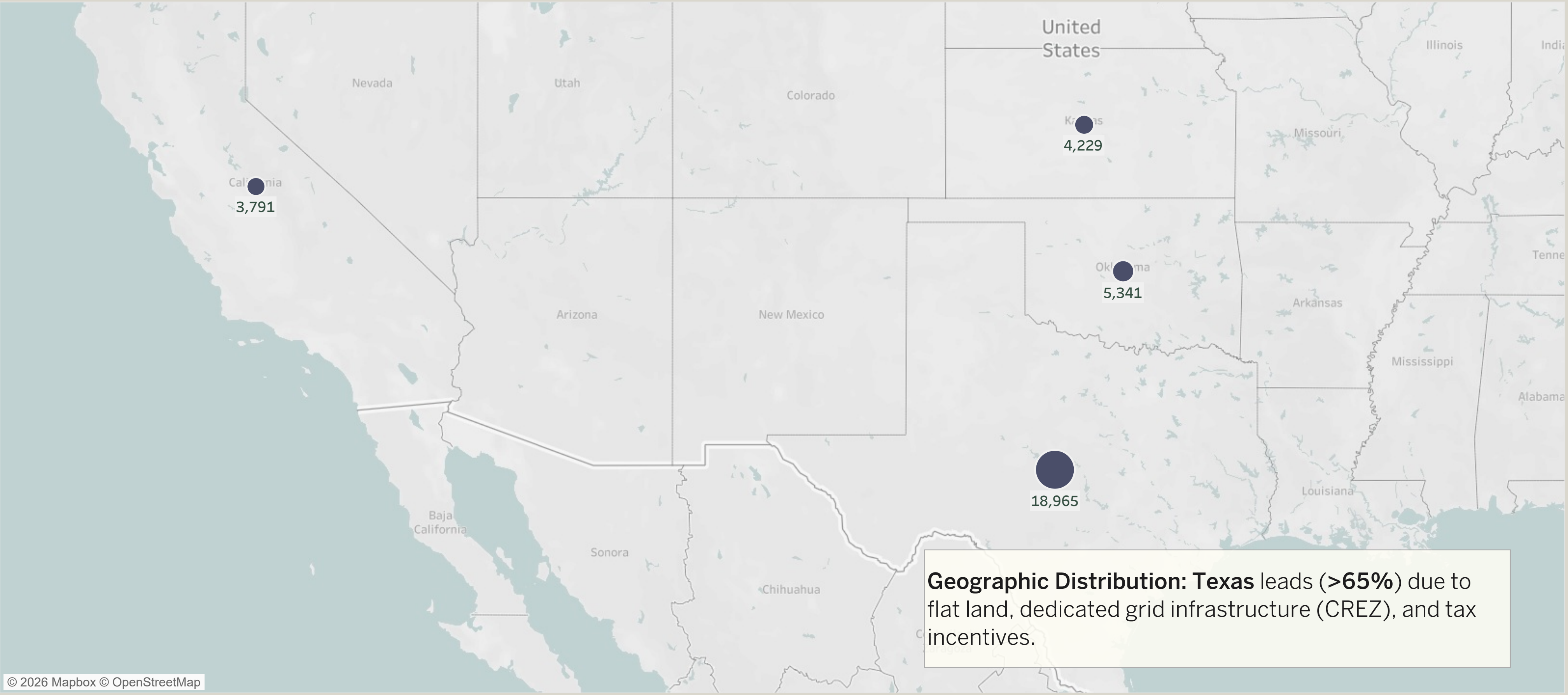
Large-scale wind projects in the US have a **high installed capacity**, showing that wind power plays an **important role in renewable electricity generation**. The country's **strong wind resources** and **large areas of suitable land** have also supported the **continued growth of wind energy**.

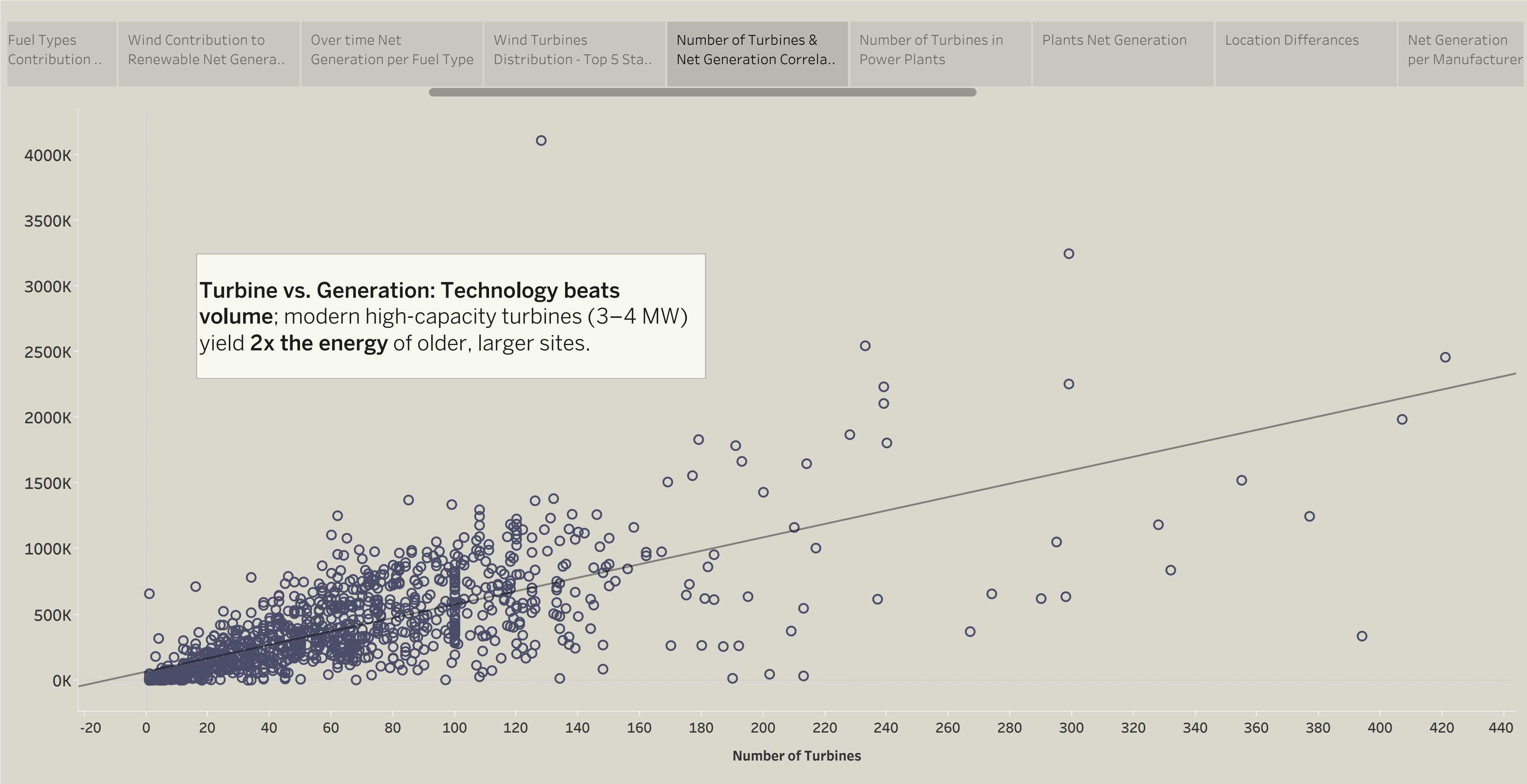
Scenario | Overview Summary | Fuel Types Contribution to Net Generation | Wind Contribution to Renewable Net Genera.. | Over time Net Generation per Fuel Type | Wind Turbines Distribution - Top 5 Sta.. | Number of Turbines & Net Generation Correla.. | Number of Turbines in Power Plants | Plants Net Generation

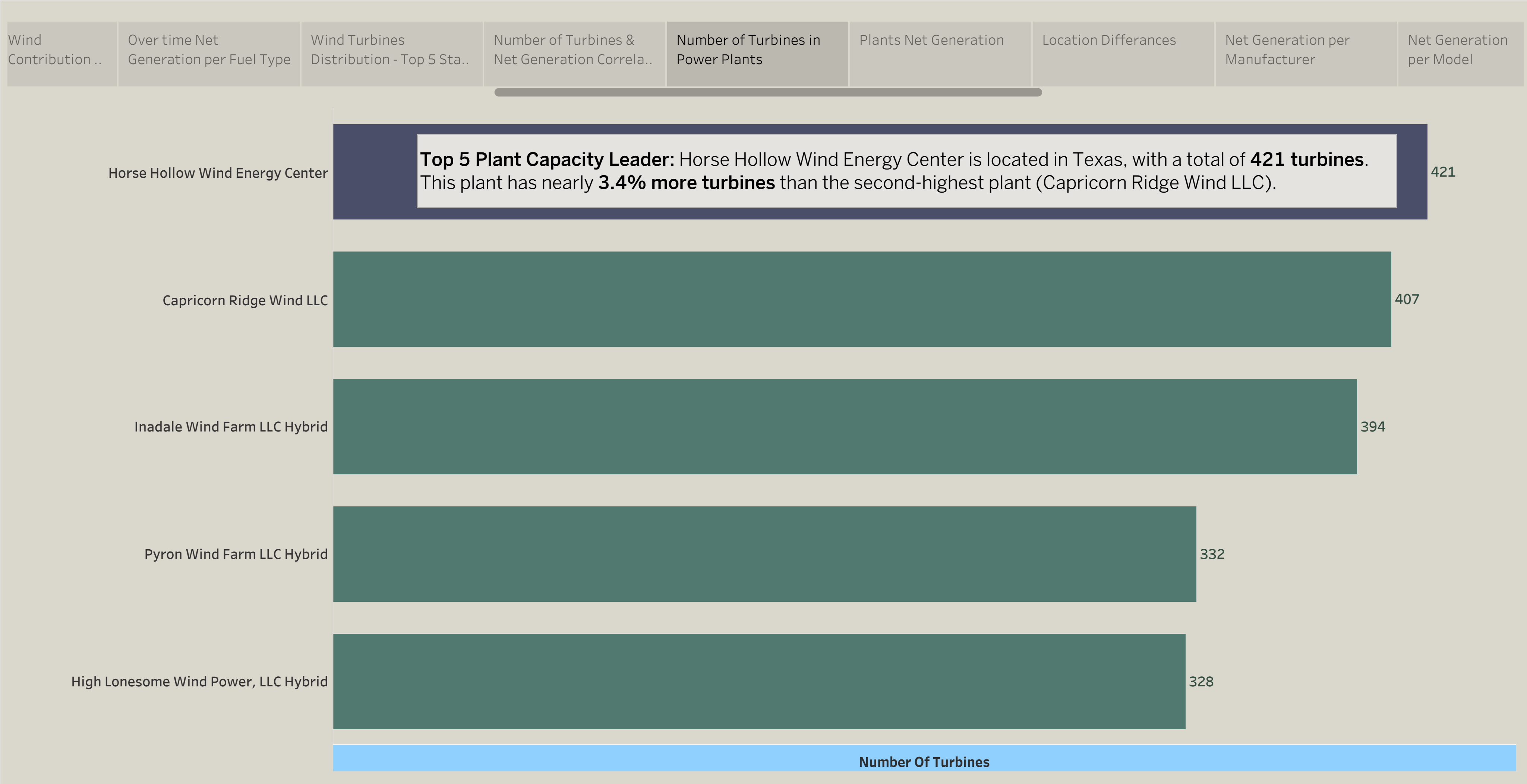


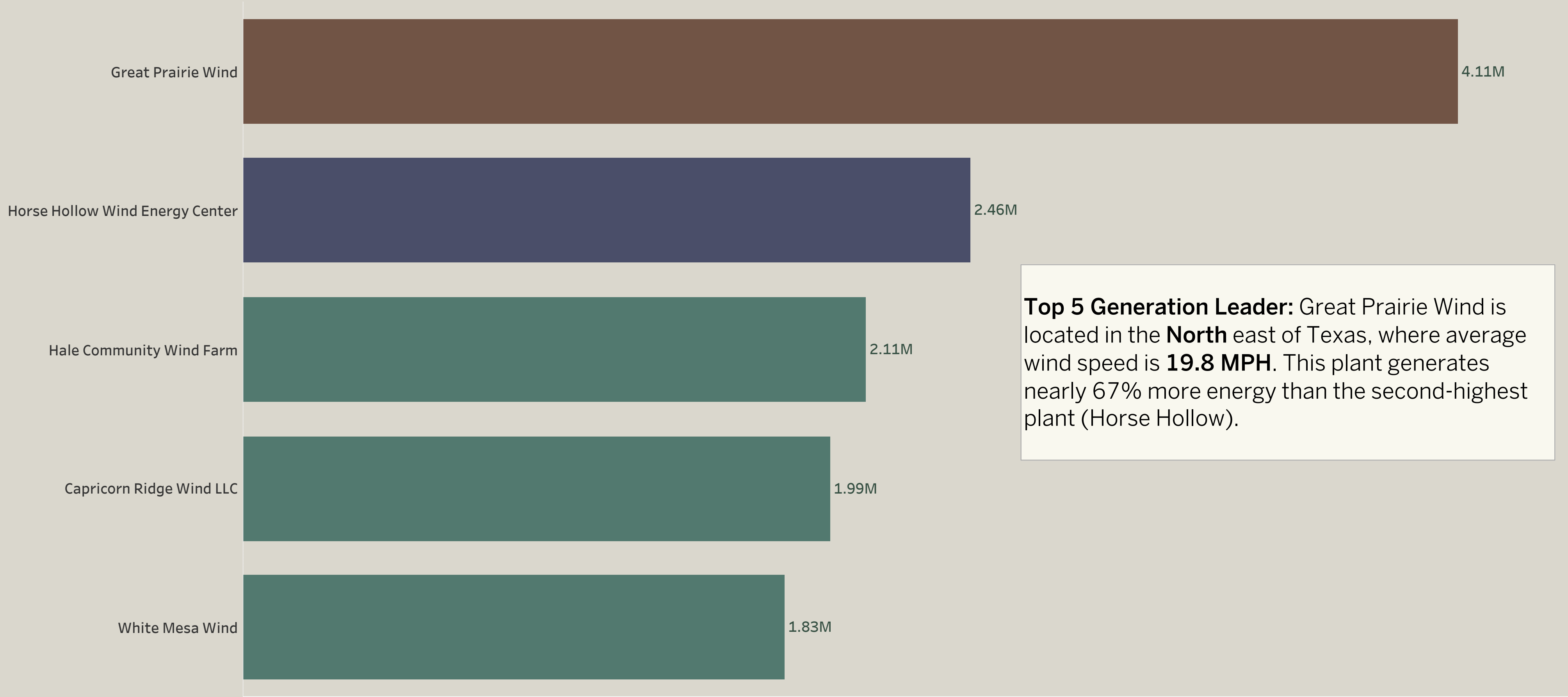
Over time Net Generation per Fuel Type's insight:
Non-renewable generation peaks at ~350M MWh in July/August and falls to ~218M MWh in April, while renewables remain relatively stable throughout the year.

Overview Summary	Fuel Types Contribution to Net Generation	Wind Contribution to Renewable Net Genera..	Over time Net Generation per Fuel Type	Wind Turbines Distribution - Top 5 Sta..	Number of Turbines & Net Generation Correla..	Number of Turbines in Power Plants	Plants Net Generation	Location Differances



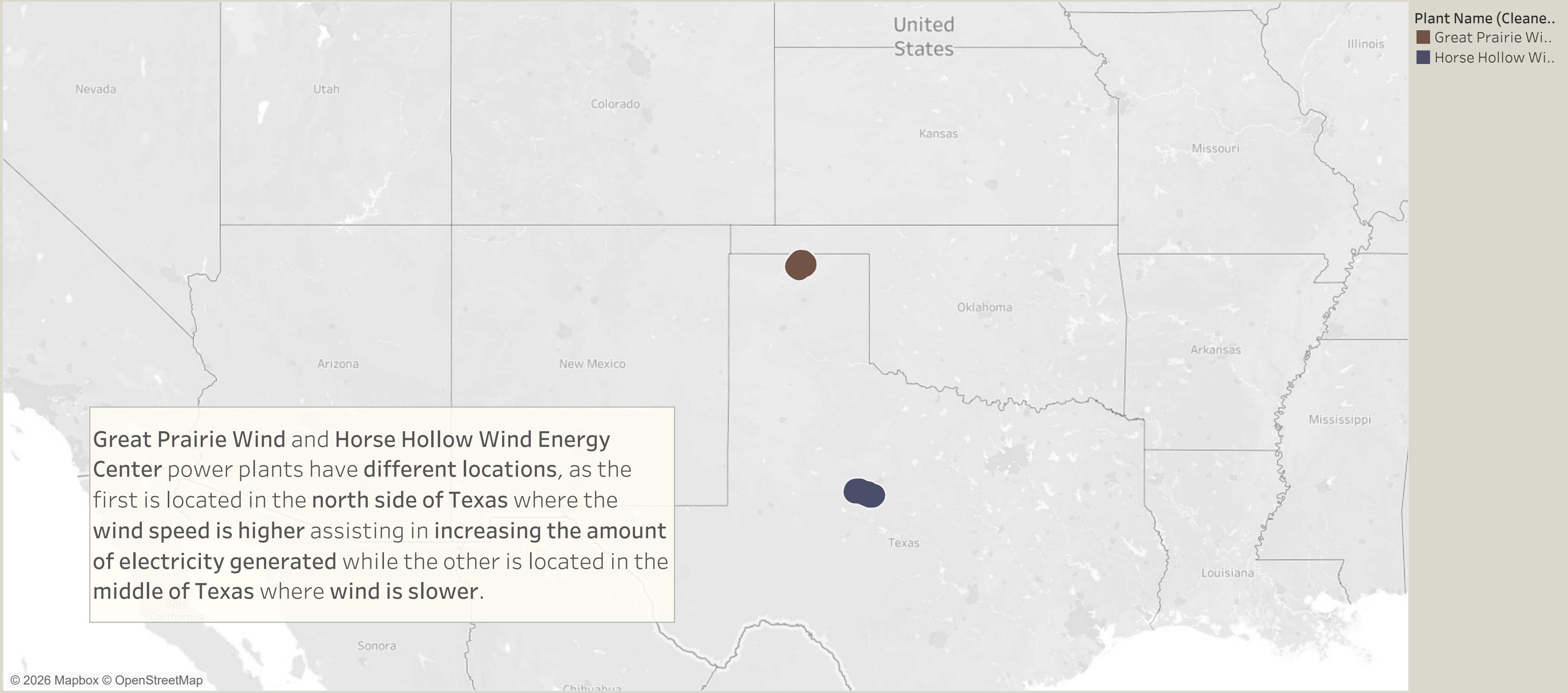

Fuel Types Contribution ..
Wind Contribution to Renewable Net Genera..
Over time Net Generation per Fuel Type
Wind Turbines Distribution - Top 5 Sta..
Number of Turbines & Net Generation Correla..
Number of Turbines in Power Plants
Plants Net Generation
Location Differances
Net Generation per Manufacturer
Turbine vs. Generation: Technology beats volume; modern high-capacity turbines (3–4 MW) yield **2x the energy** of older, larger sites.
4000K
3500K
3000K
2500K
2000K
1500K
1000K
500K
0K
-20
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
320
340
360
380
400
420
440
Number of Turbines


Wind Contribution ..
Over time Net Generation per Fuel Type
Wind Turbines Distribution - Top 5 Sta..
Number of Turbines & Net Generation Correla..
Number of Turbines in Power Plants
Plants Net Generation
Location Differances
Net Generation per Manufacturer
Net Generation per Model
Horse Hollow Wind Energy Center
Top 5 Plant Capacity Leader: Horse Hollow Wind Energy Center is located in Texas, with a total of 421 turbines. This plant has nearly 3.4% more turbines than the second-highest plant (Capricorn Ridge Wind LLC).
421
Capricorn Ridge Wind LLC
407
Inadale Wind Farm LLC Hybrid
394
Pyron Wind Farm LLC Hybrid
332
High Lonesome Wind Power, LLC Hybrid
328
Number Of Turbines

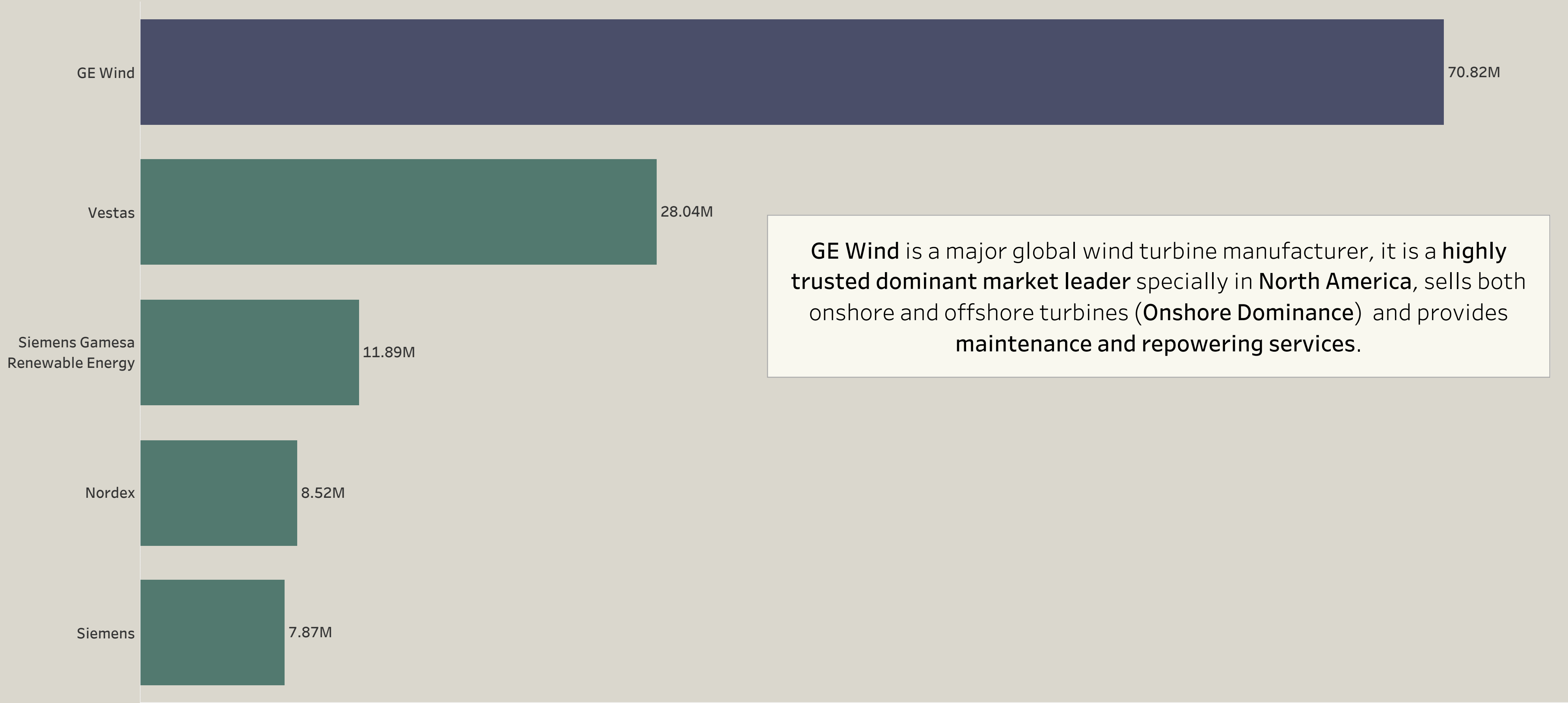


Top 5 Generation Leader: Great Prairie Wind is located in the **North** east of Texas, where average wind speed is **19.8 MPH**. This plant generates nearly 67% more energy than the second-highest plant (Horse Hollow).

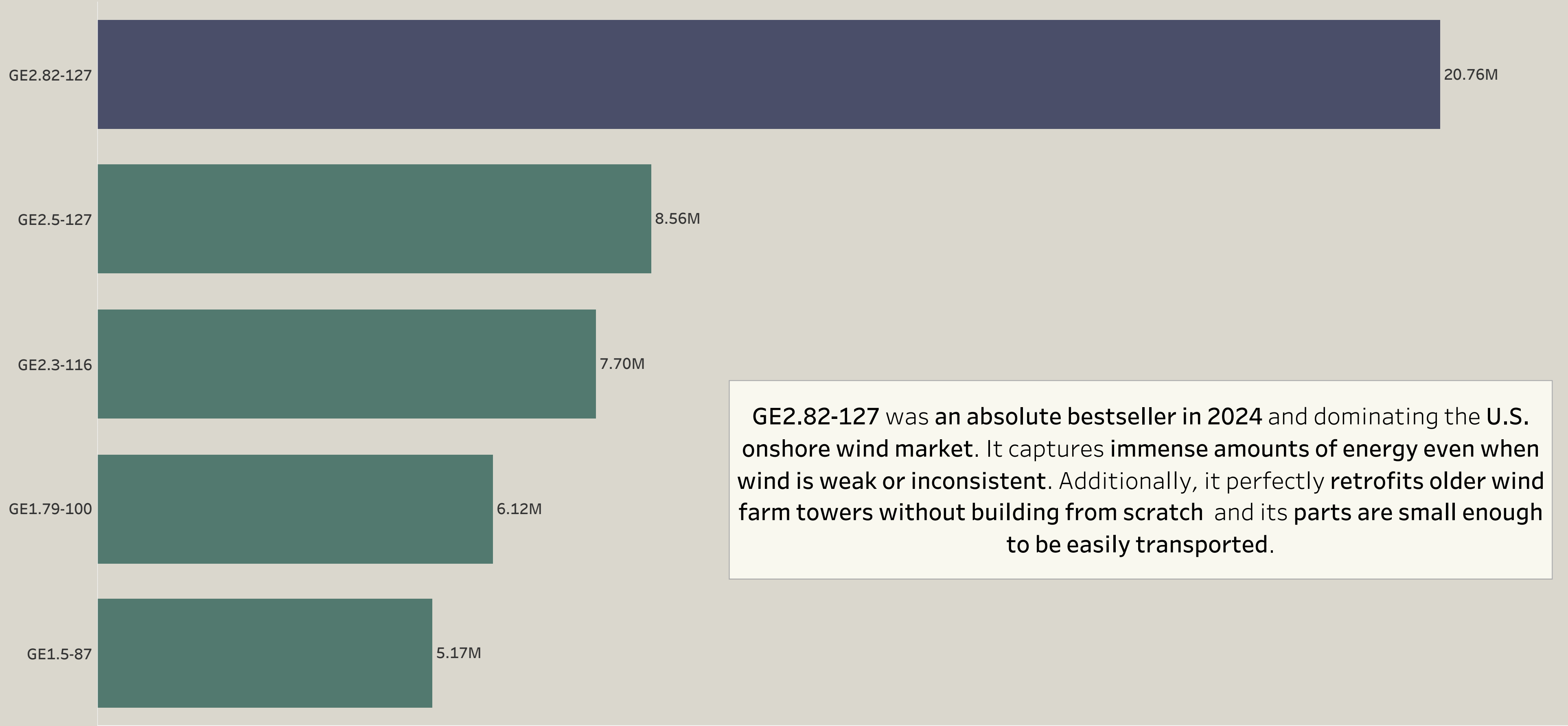
Wind Turbines Distribution - .. | Number of Turbines & Net Generation Correla.. | Number of Turbines in Power Plants | Plants Net Generation | Location Differances | Net Generation per Manufacturer | Net Generation per Model | Turbines Performance & Observed Capacity | Power Plant , Turbine Models ..



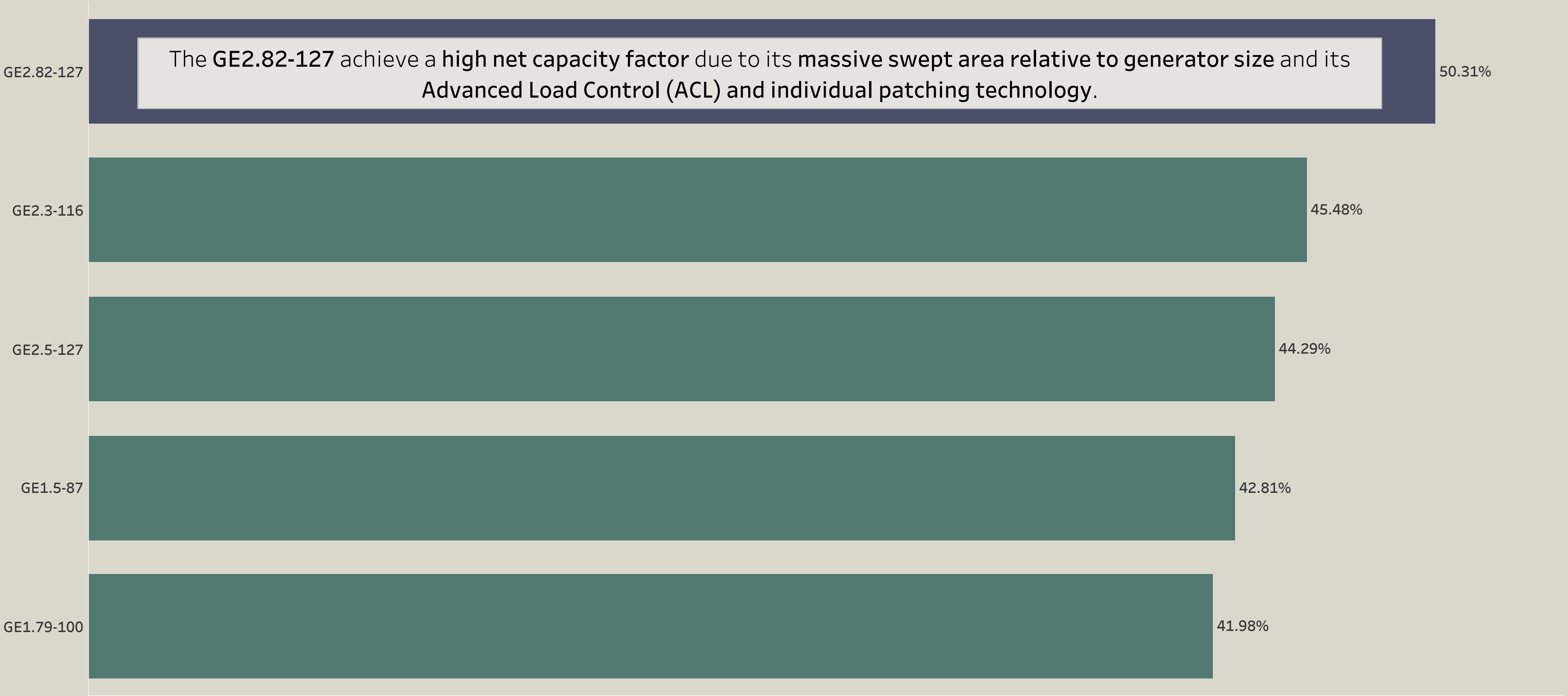
Great Prairie Wind and **Horse Hollow Wind Energy Center** power plants have **different locations**, as the first is located in the **north side of Texas** where the **wind speed is higher** assisting in **increasing the amount of electricity generated** while the other is located in the **middle of Texas** where **wind is slower**.



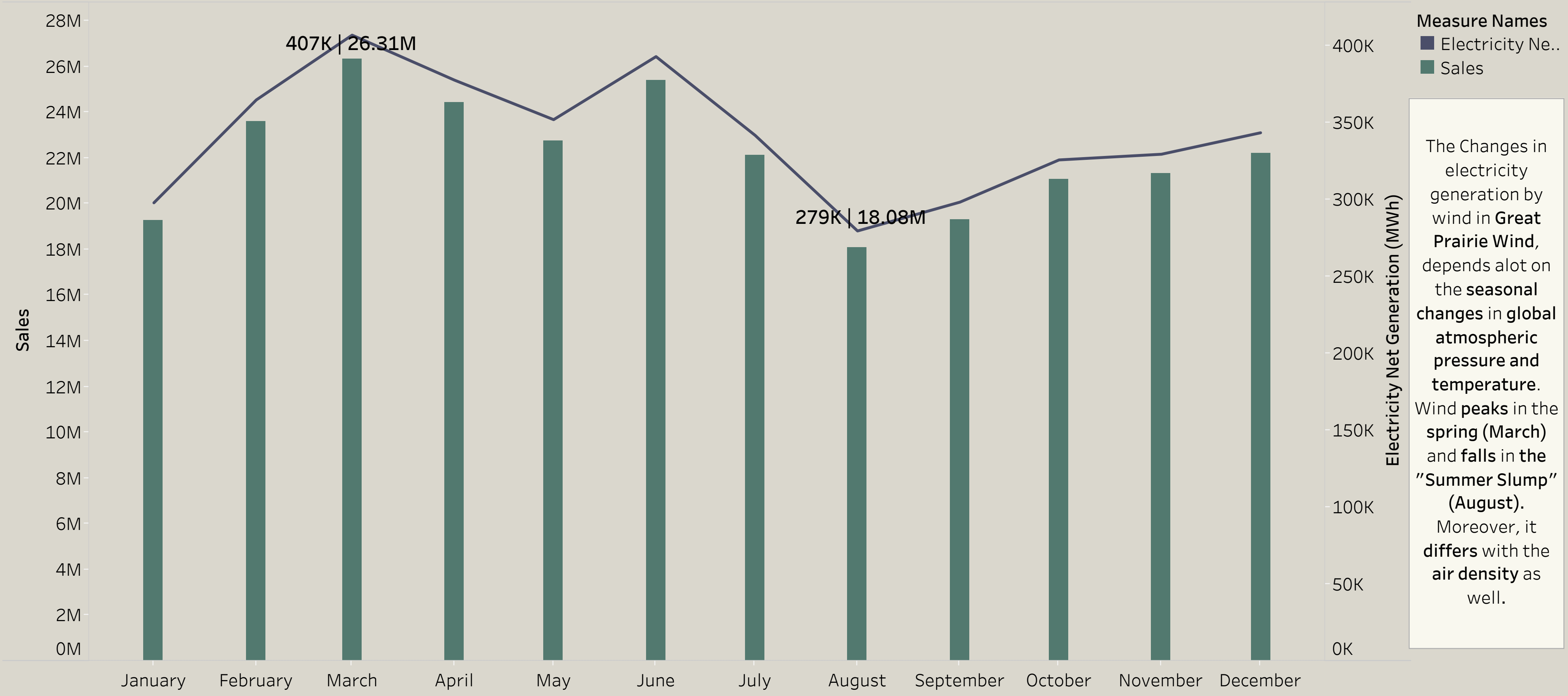
GE Wind is a major global wind turbine manufacturer, it is a **highly trusted dominant market leader** specially in **North America**, sells both onshore and offshore turbines (**Onshore Dominance**) and provides **maintenance and repowering services**.



GE2.82-127 was **an absolute bestseller in 2024** and dominating the **U.S. onshore wind market**. It captures **immense amounts of energy even when wind is weak or inconsistent**. Additionally, it perfectly **retrofits older wind farm towers without building from scratch** and its **parts are small enough to be easily transported**.


The GE2.82-127 achieve a high net capacity factor due to its massive swept area relative to generator size and its Advanced Load Control (ACL) and individual patching technology.
GE2.82-127
50.31%
GE2.3-116
45.48%
GE2.5-127
44.29%
GE1.5-87
42.81%
GE1.79-100
41.98%

Plant Name	T Manu	T Model	
Great Prairie Wind	GE Wind	GE2.82-127	Abc
Horse Hollow Wind Energy Center	GE Wind	GE1.5-87	Abc
	Siemens Gamesa Renewable Energy	SG-2.3-108	Abc



The Changes in electricity generation by wind in **Great Prairie Wind**, depends alot on the **seasonal changes** in **global atmospheric pressure and temperature**. Wind **peaks** in the **spring (March)** and **falls** in **the "Summer Slump" (August).** Moreover, it **differs** with the **air density** as well.

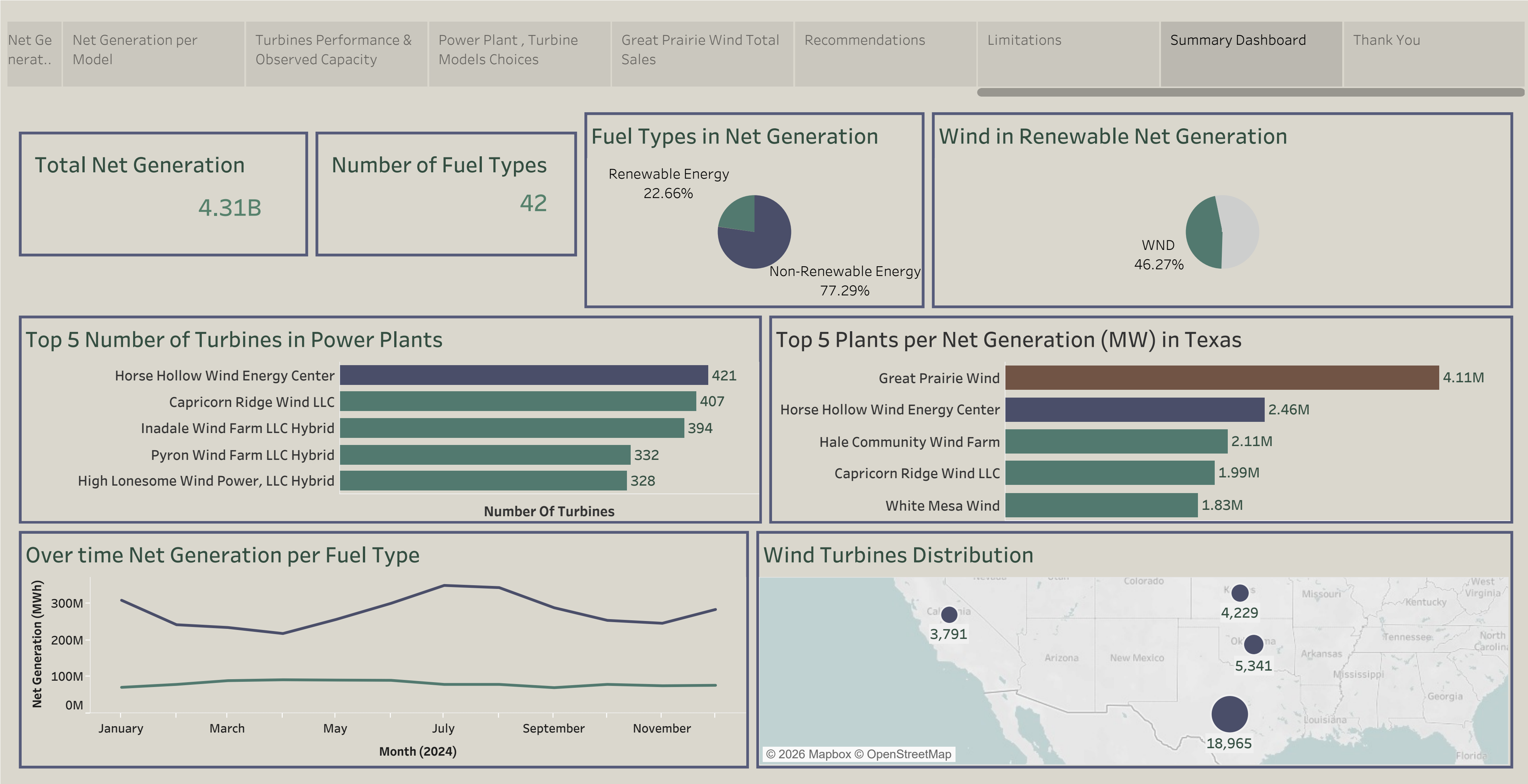
RECOMMENDATIONS

Invest in **Great Prairie Wind Power Plant**

1. Utilize advanced wind turbine **models**, such as **GE Wind (GE 2.82-127),** to ensure **higher efficiency and reliability.**
2. Select a **strategic location** in **North Texas** where **wind resources are abundant** and **turbine performance is optimal**.
3. **Scale smartly** by **adding more turbines**, while c**arefully considering** the **manufacturer and model** to **balance output with cost efficiency**.
4. Leverage **North Texas's active industrial development market** to **support long-term growth** and **maximize project impact**.

LIMITATIONS

1. We are missing the **cost information** for the **operation and maintenance,** which could have helped in analysing the **monthly cost and revenue**.
2. The dataset is **missing the amount of electricity generated** per **each turbine in the reported period** instead of the full power plant (the estimation was done assuming that all turbines worked equally).
3. No data for **downtime and maintenance frequency per turbine** was in the dataset.


Net Generat..
Net Generation per Model
Turbines Performance & Observed Capacity
Power Plant , Turbine Models Choices
Great Prairie Wind Total Sales
Recommendations
Limitations
Summary Dashboard
Thank You
Total Net Generation
4.31B
Number of Fuel Types
42
Fuel Types in Net Generation
Renewable Energy 22.66%
Non-Renewable Energy 77.29%
Wind in Renewable Net Generation
WND 46.27%
Top 5 Number of Turbines in Power Plants
Horse Hollow Wind Energy Center 421
Capricorn Ridge Wind LLC 407
Inadale Wind Farm LLC Hybrid 394
Pyron Wind Farm LLC Hybrid 332
High Lonesome Wind Power, LLC Hybrid 328
Number Of Turbines
Top 5 Plants per Net Generation (MW) in Texas
Great Prairie Wind 4.11M
Horse Hollow Wind Energy Center 2.46M
Hale Community Wind Farm 2.11M
Capricorn Ridge Wind LLC 1.99M
White Mesa Wind 1.83M
Over time Net Generation per Fuel Type
Net Generation (MWh)
300M
200M
100M
0M
January
March
May
July
September
November
Month (2024)
Wind Turbines Distribution
3,791
4,229
5,341
18,965
Arizona
New Mexico
Missouri
Kentucky
Tennessee
Arkansas
Mississippi
Georgia
Louisiana
© 2026 Mapbox © OpenStreetMap

THE END

THANK YOU